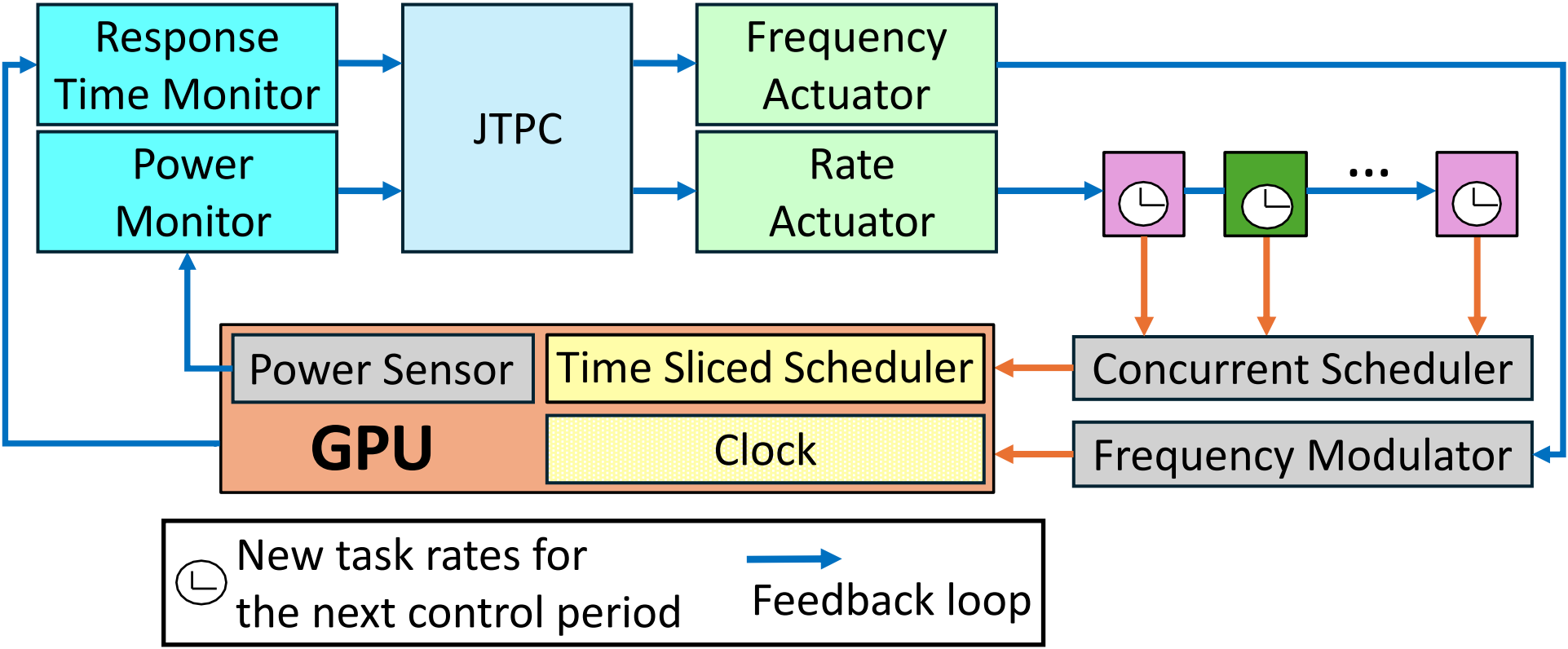$$\begin{bmatrix} r \\ f \end{bmatrix} = -\begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix} \begin{bmatrix} RT \\ P \end{bmatrix}$$